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From Geometry to a Real-Time Digital Twin of Rolling-Disc Notched Tendon-Driven Continuum Robots

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Repository: github.com/om-mpawar/ME299-25110218

Abstract. Tendon-driven continuum robots (TDCRs) bend along their whole length, which makes them useful in confined spaces but hard to design and to model. This project has two parts. **Part A** (IEEE ICRA 2027, under review, second author) asks how the geometry of a rolling-disc notched backbone sets its bending, and answers it with an automated CAD-to-FEM pipeline (616 simulations) and three printed prototypes. **Part B** (in preparation for IEEE RA-L, lead author) asks whether the whole shape of a modular robot can be predicted in real time from the servo commands alone. A beam model whose cross-section follows the CAD, with pull-only tendons and gravity, is identified once on one and two modules and predicts unseen four-module stacks and new mountings to 0.5 mm against FEM, in 1 to 4 ms per pose. The report also records what I learnt, how the approach changed with evidence, and the plan to take the work to hardware.

1 The robot, the two questions and this repository

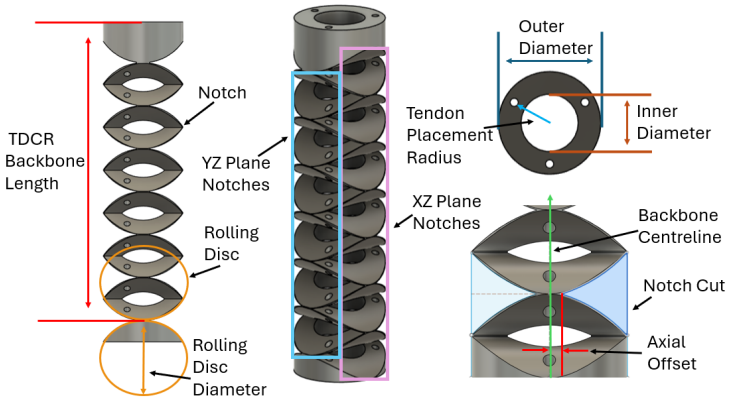


Fig. 1. The robot: a printed TPU tube with two orthogonal families of rolling-disc notches and tendons in its wall. Modules of 160 mm stack into longer robots.

A. Design. Which geometry gives a wanted bend, and what tendon stroke and force does it cost?

B. Model. Can the whole shape be predicted in real time from the servo commands alone, for any number of modules and any mounting, without sensors on the robot?

616
FEM runs in the geometry study (A)

0.50 mm
tip error, 4 unseen modules (B)

1–4 ms
per pose, 250–970 Hz (B)

132
automated tests, one-command reproduction (B)

Table 1. What is in the repository and what I did in each part.

Folder	Content	Status	My role
ICRA/	SWEEP: parametric CAD → tetrahedral mesh → SOFA FEM pipeline and the geometry study	IEEE ICRA 2027, under review. D. Yadav, **O. M. Pawar**, Krishnapriya R, S. Gandhi, M. Vadali	Second author: ported the pipeline to SOFA 25.06, parallel sweeps, mesh-fidelity and settling checks
RA-L/	**tdcr-twin**: the digital-twin package, FEM data, results, paper draft	In preparation for IEEE RA-L	Lead: model, solver, identification, baselines, software, reproduction, paper
learning/	Literature review, progress report (30 Sep), method and plan (19 Oct)	Course record	Author
poster/	A0 poster of both parts	Final evaluation	Author

2
papers: ICRA 2027 (under review), IEEE RA-L (in preparation)

12 300
lines of Python in the twin: library, scripts, tests

1 291
FEM simulations analysed (616 in A, 675 poses in B)

~30
papers read; citations checked against the publisher

2 What I learnt

2.1 Literature review

I read about thirty papers; full notes are in **learning/literature_review.md**. Table 2 lists the ones that changed what I built.

Table 2. Literature that shaped the project.

Topic	Key idea	What I took from it
Constant-curvature kinematics [1, 2]	Each section bends as a circular arc	The first baseline. It fails on a loaded notched robot: its error grows about five-fold with notch count (A) and reaches 11 to 68 mm (B)
Tendon mechanics and coupling [3, 4]	Tendon stroke = bending + axial compression; tendons of distal sections load proximal ones; slack	The core of the twin and the stroke budget of Part A. Also showed the pull-only model is not new, so the contribution moved to identification and transfer
Cosserat rods [5, 6]	Exact rod statics; displacement inputs need complementarity	Pull-only tendons as a convex problem; an independent Cosserat solver to check mine (0.004 mm)
Benchmarking [7]	Report tip error as % of robot length	The hardware pass criterion: 2 % of length, fixed before data
FEM for soft robots [8, 9, 10]	SOFA with cable constraints as ground truth	A simulator must be checked before it is trusted: I found cables that could push and gravity 1000× too weak
Shape estimation [11, 12, 13]	GP on SE(3) from sensors; observers from actuator readings; modular models of one robot	The gap: no work calibrates on 1 to 2 sections and predicts longer stacks or other mountings without sensors
Modular digital twins [14]	Calibrate modules alone, compose them	Closest work. There, actuation is local to a module; here tendons load every module below, so a module cannot be calibrated alone
Physics-informed learning [15–18]	PINNs and neural operators for continuum-robot shape	They learn the solution for one robot and are retrained per design. I learn the law inside an exact solver, and use operators across designs as the next step

2.2 How the approach changed with evidence

Early Sep
Virtual sensor
Constant curvature + FEM + a Gaussian process on the tip.
Tip-only learning says nothing about the body.

11 Sep
SOFA 25.06 port
The lab's CAD→FEM pipeline, used for the ICRA study (submitted 15 Sep).
Read logs: timeouts are not equilibria.

30 Sep
Whole-body GP
Shape modes along the backbone: tip 14.7→0.16 mm on one section.
Fixed pushing cables, weak gravity.

4–6 Oct
Literature check
The pull-only model is Camarillo 2009. Switched to the real 6-channel robot and a beam built from its CAD.
Check novelty early.

Mid Oct
Law, not residual
Learned residuals did not carry over to longer stacks; a module law does.
Learn what transfers.

Late Oct
PINN baseline, hardware
PINNs fit but do not transfer. Printable TPU modules and couplers.
Test the strongest alternative.

Early Nov
Package and audit
Tests, one-command reproduction, physics audit: torsion ×2.45, a 5th calibration number.
Every number must regenerate.

2.3 Skills gained

Mechanics
Constant curvature, Euler–Bernoulli and Timoshenko beams, Cosserat rods, tendon routing and coupling, section properties from CAD

Simulation
SOFA FEM (corotational tetrahedra, cable constraints, contact), CadQuery CAD, CGAL meshing, mesh and settling checks

Optimisation
Convex QP and LCP for pull-only tendons, fixed-point gravity, nonlinear least squares, identifiability

Learning
Gaussian processes, functional PCA shape modes, forward and inverse PINNs, cross-validation, transfer tests

Software
A Python package, 132 tests, claim-by-claim reproduction, git, LaTeX, scientific figures

Research practice
Verifying citations, positioning against prior work, stating pass criteria before data, writing for review

Record of the semester (in **learning/**): progress report to the supervisor (30 Sep), method and research plan (10 Oct, revised 19 Oct) and the literature review.

Each notch is removed from a tube of outer radius R_o and length L by a rolling disc of diameter $D_{RD} = 2R_d$; with a clearance c between cuts, the notch pitch is

$$p = D_{RD} - 2\sqrt{R_d^2 - R_o^2} + c \tag{1}$$

so D_{RD} sets how many notches fit. SWEEP turns a parameter set into a measured bend with no manual step (Fig. 2): 14 designs × 4 tendons × 11 loads = 616 runs (614 completed), TPU with the measured E = 56 MPa, gravity on. Claims are limited to tension $T \leq 0.9$ N, where the model stays valid.

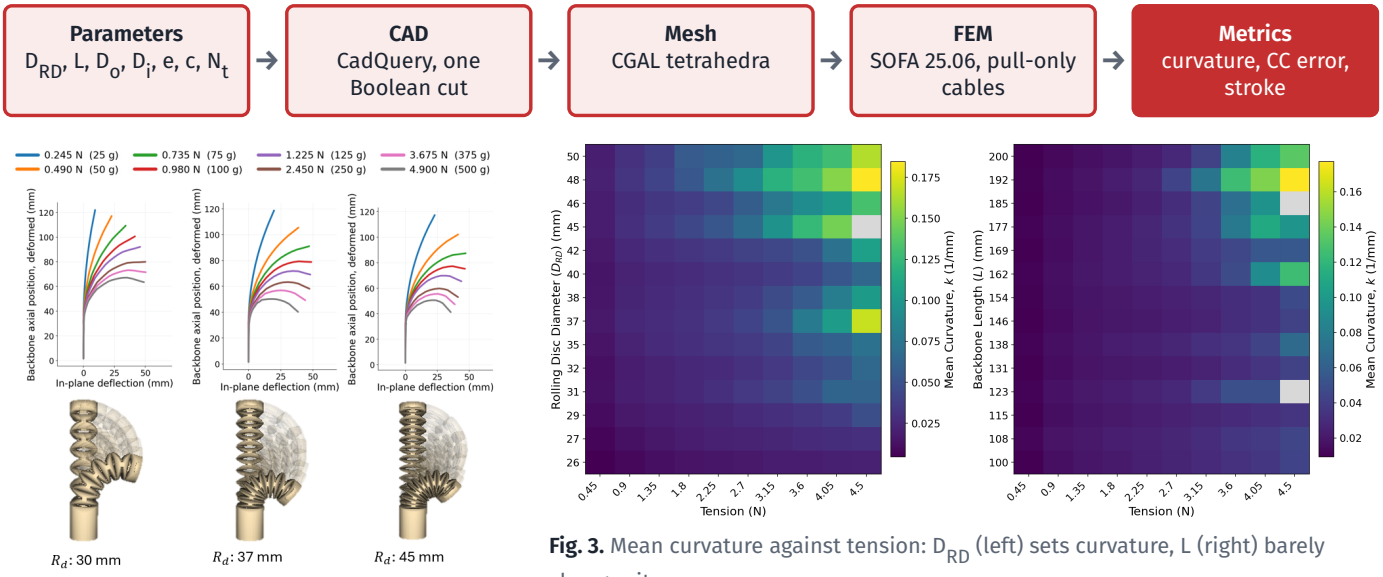


Fig. 2. Same load, three disc sizes: a larger disc gives more, shallower notches and a tighter bend.

Fig. 3. Mean curvature against tension: D_{RD} (left) sets curvature, L (right) barely changes it.

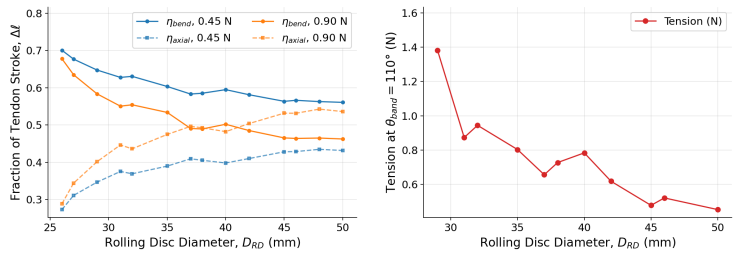


Fig. 4. Stroke budget (2): bending and compression fractions, and the tension for a 110° bend, against D_{RD}.

$$\underbrace{\Delta\ell}_{\text{tendon stroke}} = \underbrace{\Theta\, d}_{\text{bending}} + \underbrace{\Delta L}_{\text{axial compression}} \tag{2}$$

Table 3. Printed prototypes (single tendon, hanging weights, ten tracked markers): NRMSE, % of length.

D_{RD} (mm)	centreline	tip	axial
30	18.5	20.9	43.1
67.89	10.1	23.5	11.9
86.84	16.3	64.7	23.9

- **Disc diameter sets curvature.** 26 to 50 mm: 4 to 12 notches, bend 54° to 143° at 0.9 N, but the deviation from constant curvature grows about five-fold.
- **Length sets reach.** 100 to 200 mm: tip travel 78 to 157 mm while curvature changes by less than 20 %.
- **Nearly half the stroke is compression.** The budget (2) closes to 0.99 on notched designs (0.72 on a plain tube); 14 to 17 mm of a 33 to 36 mm stroke shortens the body instead of bending it. A larger disc halves the force for a 110° bend (0.87 to 0.45 N).
- **Four tendons suit the structure.** The anisotropy index $A_\kappa = \sigma_\kappa / |\vec{\kappa}|$ falls from 6.2 % to 2.6 % (13 of 14 designs), and two antagonistic pairs need two motors per section.
- **Experiments agree on shape, less on the tip** (Table 3): friction and self-contact are not modelled.

My contribution. I ported the pipeline from SOFA 23 to 25.06 (batch runner, removed components, a duplicate-mass fix, failure classification), added parallel and phased sweeps, a CAD-versus-mesh volume check and a per-run check for timeouts and self-contact, and reported the settling and mesh-convergence limits that set the 0.9 N validity range.

Input: servo commands and the mounting. Output: the backbone pose every 5 mm. Nothing is measured on the robot; one module model is composed by statics into any stack.

4.1 The physics

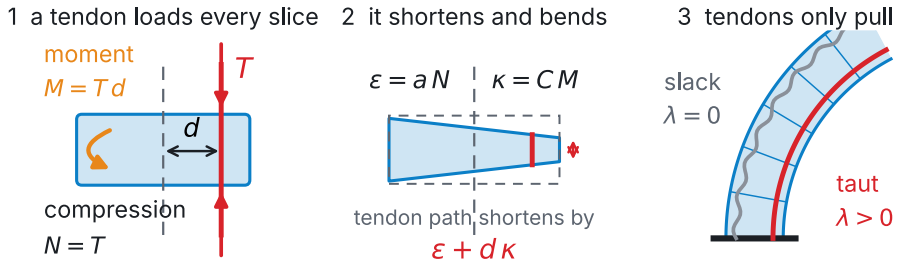


Fig. 5. (1) A tendon at offset d loads every slice it passes with a moment Td and a compression T; (2) the slice shortens and bends; (3) tendons only pull, so some go slack.

Integrating the slice compliances from the CAD gives a map from tensions to tendon shortening that is exactly linear at any bend angle, and pull-only tendons become a convex problem with a unique solution:

$$\delta\ell = G\,\lambda, \qquad G_{ij} = \sum_c \tau_{ic}\,\tau_{jc}\,\left(a_c + \boldsymbol{d}_i^{\top} C_c\,\boldsymbol{d}_j\right) + c_t L_i\,\delta_{ij}, \qquad a_c = \int_c \frac{dy}{A(y)},\;\; C_c = \int_c \frac{dy}{J(y)} \tag{3}$$

$$\lambda^{\star} = \arg\min_{\lambda \geq 0}\; \tfrac{1}{2}\,\lambda^{\top} G\,\lambda - \delta\ell^{\top}\lambda \tag{4}$$

Here $\tau_{ic} \in \{0,1\}$ marks the slices tendon i passes and $c_t L_i$ is tendon stretch. Gravity in any direction is added by a fixed point around (4). In a notch only two 4 mm bridges of the sleeve remain: area ÷8, bending stiffness ÷280, so the notches do the bending.

Verification. The solver matches closed-form beam cases to under 1 µm and an independent Cosserat rod to 0.004 mm.

4.2 Scientific ML: learn the law, not the solution

Beam theory underestimates how compliant a short notch is. I learn the law, not the shape: one scale per notch family, fitted *through* the exact solver on 675 FEM poses of one and two modules,

$$C_\theta(y) = C_{\mathrm{CAD}}(y)\,e^{\theta_k} \;\; (y \text{ in notch family } k), \qquad \theta^{\star} = \arg\min_\theta \sum_{\mathrm{poses}} \left\| x_{\mathrm{FEM}} - x_{\mathrm{twin}}(\delta\ell;\theta) \right\|^2 \tag{5}$$

giving notch compliances ×20.2 (axial) and ×5.9 / 5.3 (bending) of beam theory, and torsion ×2.45 from a twist test. The law has four physical numbers and is reused unchanged for every stack and mounting.

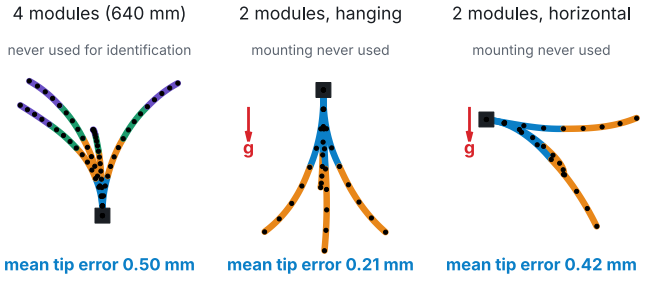


Fig. 6. Twin (colour by module) against FEM (dots) on configurations never used for identification.

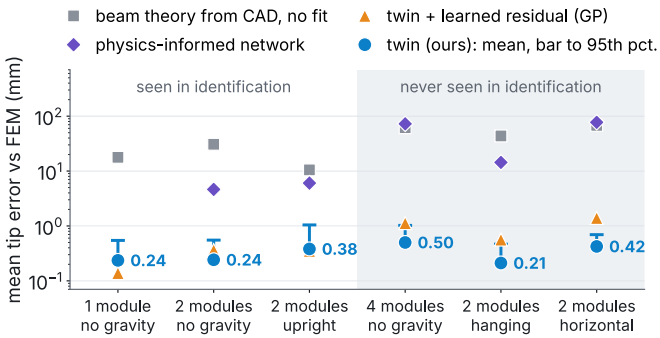


Fig. 7. Mean tip error, 60 to 100 held-out poses per set; bars: 95th percentile.

Table 4. Mean tip error against FEM (mm); 4-module length 640 mm; 1 to 4 ms per pose. **Limit:** all errors are against FEM; hardware is next (Section 5).

Model	seen (1–2 modules)	new mounting	4 modules
Beam theory from CAD, no fit	10.5–30.8	43.6–68.0	61.7
Twin + learned residual (GP)	0.14–0.36	0.56–1.40	1.12
PINNs, forward and inverse (learn the solution)	3.4–6.0	9.2–77	54–73, retrained
Identified law in the exact solver (ours)	**0.24–0.38**	**0.21–0.42**	**0.50 (0.08 % of length)**

5 Future scope: how this long-term project will be executed

The two parts meet in one direction: SWEEP generates designs (A), and the twin turns each design into a module law (B). Making that law a function of geometry gives a model that answers *"which backbone gives this shape with this stroke?"* in milliseconds.

1 Hardware validation Nov 2026 – early 2027

Print and weigh four modules; pretension 1 : 1.75 : 2.25 : 2.5. PhaseSpace motion capture. Calibrate 5 numbers (the four of (5) and the moment arm) on about 15 poses of two modules; test four modules upright and hanging.

Pass: mean tip error ≤ 2 % of length (13 mm), fixed before any data.

2 Geometry-conditioned law 2027

Identify the law for each SWEEP design and fit $\theta = f(D_{RD}, D_i, L)$, first as a smooth regression, then as a neural operator if the data demand it.

Pass: beats beam theory from the CAD on designs held out entirely.

3 Inverse design and control 2027 onwards

Target curvature and stroke budget → geometry; print and verify. Use the twin as the model in closed-loop shape control.

Pass: a printed design meets its target bend within the Phase 1 tolerance.

Table 5. Risks and the planned response.

Risk	Sign in the data	Response
Notch law not linear on hardware	Residual grows with bend angle	A convex notch energy $\psi_\theta(N, \mathbf{M})$ in place of the linear law, kept inside the same solver
Friction in the tendon channels	Hysteresis between loading and unloading	The capstan term already in the code; load in one direction only
Self-weight confounded with stiffness	Stiffness and mass trade off in the fit	Weigh every module; never fit mass from shape
Creep of TPU	Drift at constant command	Fixed settling time per pose; repeat poses at the start and end

Outputs planned. ICRA 2027 decision and revision (early 2027); RA-L submission once Phase 1 gives a hardware table; the twin as an installable package.

6 Conclusions and what I gained

- **Design (A):** disc diameter sets curvature, length sets reach, and nearly half the tendon stroke goes into compression; four tendons suit the two-plane symmetry.
- **Model (B):** a module law identified on short stacks, inside exact physics, predicts longer stacks and new mountings to 0.5 mm against FEM in milliseconds; learned solutions do not transfer.
- **What I gained:** continuum-robot mechanics from first principles, FEM practice and its pitfalls, convex optimisation and identification, where learning helps and where physics must carry the generalisation, and research habits: check novelty, verify the simulator, state pass criteria before data, make every number reproducible.

References

1. R. J. Webster, B. A. Jones, "Design and kinematic modeling of constant curvature continuum robots: a review," IJRR, 29(13), 2010.
2. J. Burgner-Kahrs, D. C. Rucker, H. Choset, "Continuum robots for medical applications: a survey," IEEE T-RO, 31(6), 2015.
3. D. B. Camarillo et al., "Mechanics modeling of tendon-driven continuum manipulators," IEEE T-RO, 24(6), 2008.
4. D. B. Camarillo, C. R. Carlson, J. K. Salisbury, "Configuration tracking for continuum manipulators with coupled tendon drive," IEEE T-RO, 25(4), 2009.
5. D. C. Rucker, R. J. Webster, "Statics and dynamics of continuum robots with general tendon routing and external loading," IEEE T-RO, 27(6), 2011.
6. K. Oliver-Butler, J. Till, C. Rucker, "Continuum robot stiffness under external loads and prescribed tendon displacements," IEEE T-RO, 35(2), 2019.
7. P. Rao, Q. Peyron, S. Lilge, J. Burgner-Kahrs, "How to model tendon-driven continuum robots and benchmark modelling performance," Front. Robot. AI, 7, 2020.
8. F. Faure et al., "SOFA: a multi-model framework for interactive physical simulation," Springer, 2012.
9. E. Coevoet et al., "Software toolkit for modeling, simulation, and control of soft robots," Adv. Robot., 31(22), 2017.
10. C. Duriez, "Control of elastic soft robots based on real-time finite element method," ICRA, 2013.
11. S. Lilge, T. D. Barfoot, J. Burgner-Kahrs, "Continuum robot state estimation using Gaussian process regression on SE(3)," IJRR, 41(13–14), 2022.
12. D. Feliu-Talegon et al., "Dynamic shape estimation of tendon-driven soft manipulators via actuation readings," IEEE RA-L, 10(1), 2025.
13. Zhang et al., "A general modular framework for modeling cable-driven continuum surgical robots," Int. J. Mech. Sci., 318, 2026.
14. S. H. Kim et al., "Digital twins for the design, interactive control, and deployment of modular, fiber-reinforced soft continuum arms," arXiv: 2507.10121, 2025.
15. N. Feizi et al., "Few-shot physics-informed neural network for shape reconstruction of concentric-tube robots," ICRA, 2026.
16. R. Danesh, P. Qarehdaghi, F. Janabi-Sharifi, "Constraint-aware physics-informed neural networks for static shape estimation of co-manipulative continuum robots," arXiv:2608.26273, 2026.
17. G. Zhan et al., "Physics informed neural pose estimation for real-time shape reconstruction of soft continuum robots," IEEE RA-L, 10(7), 2025.
18. B. Frieden, J. M. Ferguson, A. Kuntz, V. Shankar, "Neural operators for design-space surrogate modeling of tendon-actuated continuum robots," ICRA, 2026.